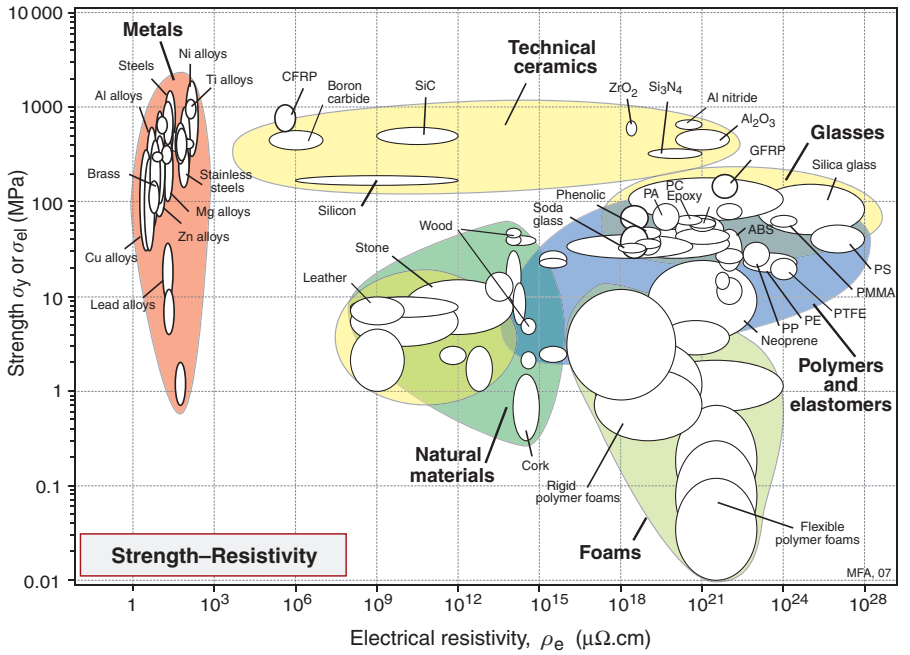


Figure 14.6 Material property chart for electrical resistivity ρ_e and yield strength σ_y or elastic limit σ_{el}.

Strength and electrical resistivity

The second chart (Figure 14.6) shows electrical resistivity ρ_e and strength σ_{el}. Metals lie at the extreme left: copper and silver (not shown) have the lowest resistivities, with aluminum only just above. The strongest alloys have resistivities that are 10–100 times larger. Ceramics such as alumina, aluminum nitride and silica, at the upper right, have very high resistivities and are strong; they are used for making insulators for power lines and substrates for electronic circuitry. Below them lie polymers that are less strong, but are flexible, easily molded and excellent insulators; they are used for cable sheathing, insulation of switch gear and the like.

Strength and dielectric loss

The final chart (Figure 14.7) shows dielectric loss factor L and strength σ_y or σ_{el}. Materials that are transparent to microwaves (radar uses microwave frequencies) lie at the left: polymer foams, certain polymers (PP, PE, PTFE) and some ceramics have particularly low values. Materials that absorb microwaves, becoming hot in the process, lie at the right: polymers containing a polar group, like nylon and polyurethane, and natural materials like woods (woods are dried by microwave heating). Water, which has a polar molecule,

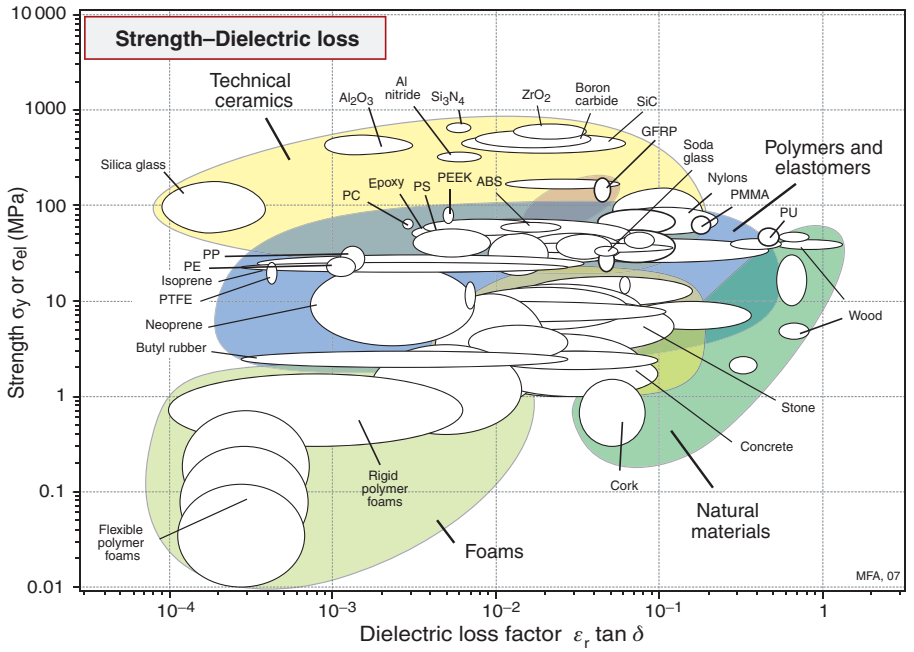


Figure 14.7 Material property chart for dielectric loss, $\varepsilon_r \tan\delta$ and yield strength σ_y or elastic limit σ_{el}.

has a particularly large loss factor, which is why most foods can be cooked in a microwave oven.

14.4 Drilling down: the origins and manipulation of electrical properties

Electrical conductivity

An electric field, E (volts/m), exerts a force Ee on a particle carrying a charge. Solids are made up of atoms containing electrons that carry a charge $-e$ and a nucleus containing protons, each with a positive charge $+e$. If charge carriers can move, the force Ee causes them to flow through the material—that is, it conducts. Metals are *electron conductors*, meaning that the charge carriers are the electrons. In ionic solids (which are composed of negatively and positively charged ions like Na^+ Cl^-) the diffusive motion of ions allows *ionic conduction*, but this is only possible at temperatures at which diffusion is rapid. Many materials have no mobile electrons, and at room temperature they are too cold

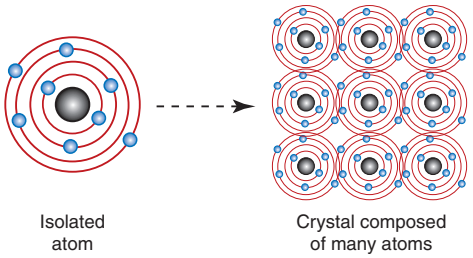


Figure 14.8 When atoms are brought together to form crystals, the outermost electrons interact and the discrete energy levels split into bands.

to be ionic conductors. The charged particles they contain still feel a force in an electric field and it is enough to displace the charges slightly, but they are unable to move more than a tiny fraction of the atom spacing. These are insulators; the small displacement of charge gives them dielectric properties.

How is it that some materials have mobile electrons and some do not? To explain this we need two of the stranger results of quantum mechanics. Briefly, the electrons of an atom occupy discrete energy states or orbits, arranged in shells (designated 1, 2, 3, etc. from the innermost to the outermost); each shell is made up of sub-shells (designated s, p, d and f), each of which contains one, three, five or seven orbits respectively. The electrons fill the shells with the lowest energy, two electrons of opposite spin in each orbit; the *Pauli*[6] *exclusion principle* prohibits an energy state with more than two. When n atoms (a large number) are brought together to form a solid, the inner electrons remain the property of the atom on which they started, but the outer ones interact (Figure 14.8). Each atom now sits in the field created by the charges of its neighbors. This has the effect of decreasing slightly the energy levels of electrons spinning in a direction favored by the field of its neighbors and raising that of those with spins in the opposite direction, splitting each energy level. Thus, the discrete levels of an isolated atom broaden, in the solid, into *bands* of very closely spaced levels. The number of electrons per atom that have to be accommodated depends only on the atomic number of the atoms. These electrons fill the bands from the bottom, lowest energy, slot on up, until all are on board, so to speak. The topmost filled energy level is called the Fermi[7] level (more on this later). An electron in this level still has an energy that is lower than it would have if it

[6] Wolfgang Joseph Pauli (1900–1958), quantum theorist, conceiver of the neutrino and Nobel Prize winner. He was not, however, a happy man, requiring psychotherapy, which he received from none other than the great psychoanalyst Carl Jung.

[7] Enrico Fermi (1901–1954), devisor of the statistical laws known as Fermi statistics governing the behavior of electrons in solids. He was one of the leaders of the team of physicists on the Manhattan Project for the development of the atomic bomb.

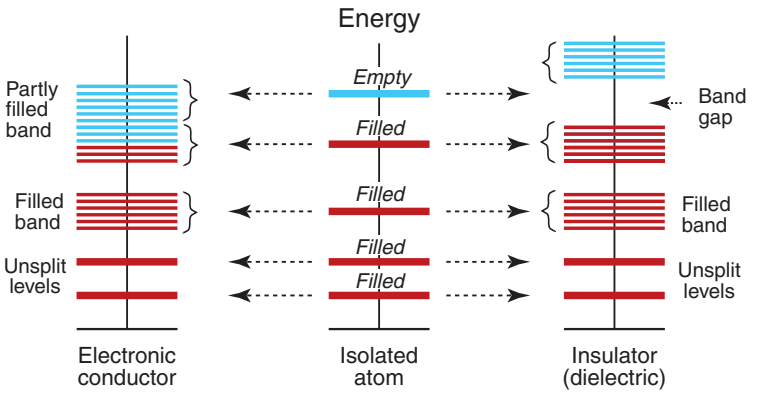


Figure 14.9 Conductors, on the left, have a partly filled outer band—electrons in it can move easily. Insulators, on the right, have an outer filled band, separated from the nearest unfilled band by a band gap. (Red bands are filled, blue bands are empty.)

were isolated in a vacuum far from the atoms. This energy difference is called, for historical reasons, the *work function* because it is the work that you have to do to remove an electron from the Fermi level to infinity. If you want to create an electron beam by pulling electrons out of a metal by using a high electric field (called a *field-emission electron gun*) you have to provide the work function to do it.

Whether the material is a conductor or an insulator depends on how full the bands are, and whether or not they overlap. In Figure 14.9 the central column describes an isolated atom and the outer ones illustrate the possibilities created by bringing atoms together into an array, with the energies spread into energy bands. Conductors like copper, shown on the left, have an unfilled outer band; there are many very closely spaced levels just above the last full one, and—when accelerated by a field—electrons can use these levels to move freely through the material. In insulators, shown on the right, the outermost band with electrons in it is full, and the nearest empty band is separated from it in energy by a wide *band gap*. Semiconductors, too, have a band gap, but it is narrower—narrow enough that thermal energy can pop a few electrons into the empty band, where they conduct. Deliberate doping (adding trace levels of impurities) creates new levels in the band gap, reducing the energy barrier to entering the empty states and thus allowing more carriers to become mobile.

Electrical resistance

If a field E exerts a force Ee on an electron, why does it not accelerate forever, giving a current that continuously increases with time? This is not what happens; instead, switching on a field causes a current that almost immediately

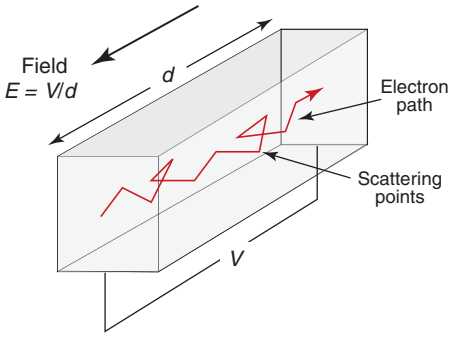


Figure 14.10 An electron, accelerated by the field *E*, is scattered by imperfection that create a resistance to its motion.

reaches a steady value. Referring back to equations (14.1) and (14.2), the current density i/A is proportional to the field E:

$$\frac{i}{A} = \frac{E}{\rho_e} = \kappa_e E \tag{14.14}$$

where ρ_e is the resistivity and κ_e, its reciprocal, is the electrical conductivity.

Broadly speaking, the picture is this. Conduction electrons are free to move through the solid. Their thermal energy $k_B T$ (k_B = Boltzmann's constant, T = absolute temperature) causes them to move like gas atoms in all directions. In doing this they collide with *scattering centers*, bouncing off in a new direction. Impurity or solute atoms are particularly effective scattering centers (which is why alloys always have a higher resistivity than pure metals), but electrons are scattered also by imperfections such as dislocations and by the thermal vibration of the atoms themselves. When there is no field, there is no *net* transfer of charge in any direction even though all the conduction electrons are moving freely. A field imposes a drift velocity $v_d = \mu_e E$ on the electrons, where μ_e is the electron mobility and it is this that gives the current (Figure 14.10). The drift velocity is small compared with the thermal velocity; it is like a breeze in air—the thermal motion of the air molecules is far greater than the 'drift' that we feel as the breeze. The greater the number of scattering centers, the shorter is the mean free path, λ_{mfp}, of the electrons between collisions, and the slower, on average, they move. Just as with thermal conductivity, the electrical conductivity depends on mean free path, on the density of carriers (the number n_v of mobile electrons per unit volume) and the charge they carry. Thus, the current density, i/A, is given by

$$\frac{i}{A} = n_v e v_d = n_v e \mu_e E$$

Comparing this with equation (14.14) gives the conductivity:

$$\kappa_e = n_v \, e \, \mu_e \tag{14.15}$$

Thus the conductivity is proportional to the density of free electrons and to the electron mobility, and this is directly proportional to the mean free path. The purer and more perfect the conductor, the higher is the conductivity.

Metals are strengthened (Chapter 7) by solid solution hardening, work hardening or precipitation hardening. All of these change the resistivity too. By 'zooming in' on part of the strength–resistivity chart, we can see how both properties are affected—Figure 14.11 shows this for two of the best conductors: copper and aluminum. Adding solute to either metal increases its solute atoms also act as scattering centers, increasing the electrical resistivity too. Dislocations add strength (by what we called *work hardening*) and they too scatter electrons a little, though not as much as solute. Precipitates offer the greatest gain in strength; their effect on resistivity varies, depending on their size and spacing compared to the electron mean free path and on the amount of residual solute left in the lattice. Precipitation hardening (with low residual solute) or work hardening are therefore the best ways to strengthen conductors. The two figures show that commercial conductor alloys have much greater strength, and only slightly greater resistivity than the pure metals.

The resistivity of metals increases with temperature because thermal vibration scatters electrons. Resistance decreases as temperature falls, which is why very-high-powered electromagnets are pre-cooled in liquid nitrogen. As absolute zero

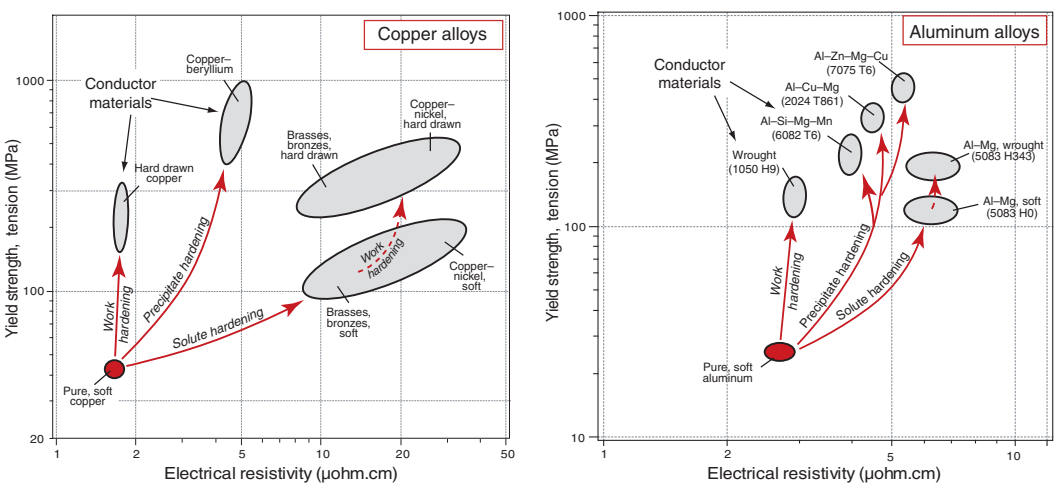


Figure 14.11 The best choice of material for a cable is one with high strength and low resistivity, but strengthening mechanisms increase resistivity. Work hardening and precipitation hardening do so less than solute hardening.

is approached most metals retain some resistivity, but a few, at between 0 and 4 K, suddenly lose all resistance and become superconducting. The resistivity of semiconductors, by contrast, decreases as temperature increases because, as explained earlier, thermal energy allows more carriers to cross the band gap.

Dielectric behavior

Dielectrics are insulators. So what happens to their electrons when a field E is applied? In zero field, the electrons and protons in most dielectrics are symmetrically distributed, so the material carries no net or dipole moment. A field exerts a force on a charge, pushing positive charges in the direction of the field and negative charges in the opposite direction. The effect is easiest to see in ionic crystals, since here neighboring ions carry opposite charges, as on the left of Figure 14.12. Switch on the field and the positive ions (charge $+q$) are pulled in the field direction, the negative ones (charge $-q$) in the reverse, until the restoring force of the inter-atomic bonds just balances the force due to the field at a displacement of Δx, as on the right of the figure. Two charges $\pm q$ separated by a distance Δx create a dipole with dipole moment, d, given by

$$d = q\Delta x \tag{14.16}$$

The polarization of the material, P, is the volume-average of all the dipole moments it contains:

$$P = \frac{\sum d}{\text{Volume}}$$

Even in materials that are not ionic, like silicon, a field produces a dipole moment because the nucleus of each atom is displaced a tiny distance in the direction of the field and its surrounding electrons are displaced in the opposite

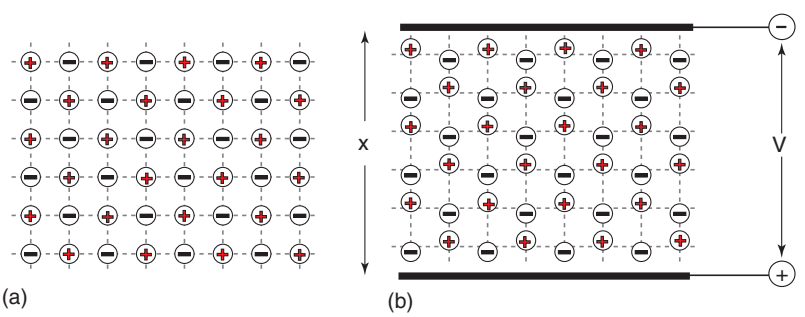


Figure 14.12 An ionic crystal in zero applied field, (a) and when a field V/x is applied (b). The electric field displaces charge, causing the material to acquire a dipole moment.

direction. The resulting dipole moment depends on the magnitude of the displacement and the number of charges per unit volume, and it is this that determines the dielectric constant. The bigger the shift, the bigger the dielectric constant. Thus, compounds with ionic bonds and polymers that contain polar groups like $-OH^-$ and $-NH^-$ (nylon, for example) have larger dielectric constants than those that do not.

Dielectric loss

Think now of polarization in an alternating electric field. When the upper plate of Figure 14.12(b) is negative, the displacements are in the direction shown in the figure. When its polarity is reversed, it is the negative ions that are displaced upwards, the positive ions downwards; in an oscillating field, the ions oscillate. If their oscillations were exactly in phase with the field, no energy would be lost, but this is never exactly true, and often the phase shift is considerable. Materials with high dielectric loss usually contain awkwardly shaped molecules that themselves have a dipole moment—the water molecule is an example (Figure 14.13). These respond to the oscillating field by rotating, but because of their shape they interfere with each other (you could think of it as molecular friction) and this dissipates energy that appears as heat—that is how microwave heating works. As equation (14.13) showed, the energy that is dissipated depends on the frequency of the electric field; generally speaking, the higher the frequency, the greater the power dissipated (because power is work per second, and the more times the molecules shuttle, the more energy is lost), but there are peaks at certain frequencies that are characteristic of the material structure.

Dielectric breakdown

In metals, as we have seen, even the smallest field causes electrons to flow. In insulators they can't, because of the band gap. But if, at some weak spot, one electron is torn free from its parent atom, the force Ee exerted by the field E accelerates it, giving it kinetic energy; it continues to accelerate until it collides with another atom. A sufficiently large field can give the electron so much kinetic energy that, in the collision, it kicks one or more new electrons out of the atom it hits, and they, in turn, are accelerated and gain energy. The result, sketched in Figure 14.14, is a cascade—an avalanche of charge. It is sufficiently violent that it can damage the material permanently.

The critical field strength to make this happen, called the *breakdown potential*, is hard to calculate: it is that at which the first electron breaks free at the

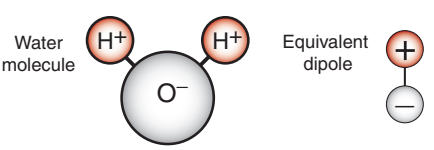


Figure 14.13 A water molecule. The asymmetry gives it a dipole moment.

weak spot—a defect in the material such as a tiny crack, void or inclusion that locally concentrates the field. The necessary fields are large, typically 1–15 MV/m. That sounds a lot, but such fields are found in two very different circumstances: when the voltage is very high or the distances are very small. In power transmission, the voltages are sufficiently high—20 000 volts or so—that breakdown can occur at the insulators that support the line, while in microcircuits and thin-film devices the distances between components are very small: a 1 V difference across a distance of 1 μm gives a field of 1 MV/m.

Piezo-electric materials

The word 'Quartz' on the face of your watch carries the assurance that its time-steps are set by the oscillations of a piezo-electric quartz crystal. Piezo-electric behavior is found in crystals in which the ions are unsymmetrically distributed (the crystal structure lacks a center of symmetry), so that each molecule carries a permanent dipole moment (Figure 14.15(a)). If you cut a little cube from such

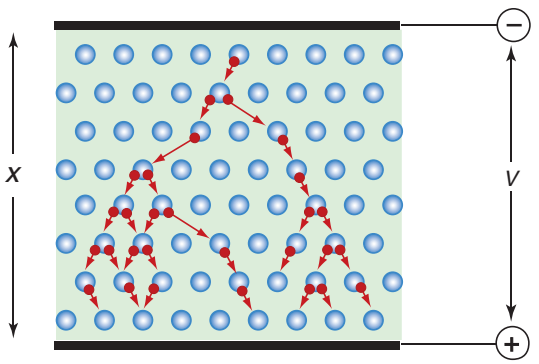


Figure 14.14 Breakdown involves a cascade of electrons like a lightning strike.

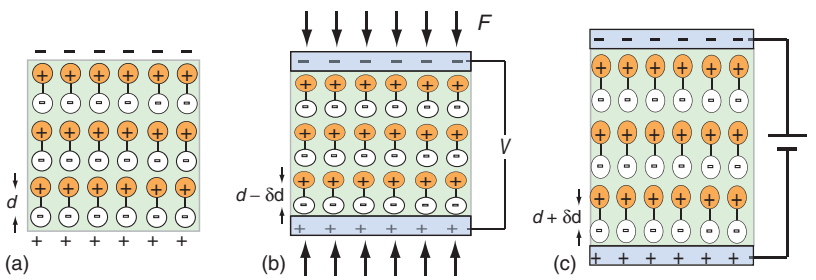


Figure 14.15 A piezo-electric material has unsymmetrically distributed charge, giving it a natural dipole moment. The surface charge associated with this is neutralized by pick-up of ions, but if it is deformed, as in (b), the dipole moment changes and the surfaces become charged. The inverse is also true: a field induces a change of shape, the basis of piezo-electric actuation, as in (c).

a material, with faces parallel to crystal planes, the faces would carry charge. This charge attracts ions and charged particles from the atmosphere just as a television screen does, giving an invisible surface layer that neutralizes the charge. If you now squeeze the crystal, its ions move relative to each other, the dipole moment changes and the charge on the surface changes too (Figure 14.15(b)). Given time, the newly appeared charge would attract neutralizing ions, but this does not happen immediately, giving a potential difference. This provides the basis of operation of electric microphones and pick-ups. The potential difference between the faces can be large—large enough to generate a spark across a narrow gap—it is the way that gas lighters work.

A strain, then, induces an electric field in a piezo-electric material. The inverse is also true: a field induces a strain. The field pulls the positive ions and pushes the negative ones, changing their spacing and so changing the shape of the crystal (Figure 14.15(c)). If a small strain produces a large field, then a large field will produce only a very small strain. But the strain is a linear function of field, allowing extremely precise, if small, displacements, used for positioning and actuation at the sub-micron scale.

Piezo-electric materials respond to a change in electric field faster than most materials respond to a stimulus of this or any other kind. Put them in a megahertz field and they respond with microsecond precision. That opens up many applications, some described later in this chapter. In particular, it opens up the world of ultrasonics—sound waves with frequencies starting at the upper limit of the human ear, 20 kHz, on up to 20 000 kHz and above.

Pyro-electric materials

Pyro-electricity—polarization caused by change of temperature—was known to (but not understood by) the Greek philosopher Theophrastus,[8] who noted that certain stone, when warmed or cooled, acquired the ability to pick up straw and dry leaves. It was not until the 20th century that this strange behavior was understood and exploited.

Some materials have a permanent dipole moment because their positive and negative ions balance electrically but are slightly out of line with each other. If a thin disk of one of these is cut so that its faces are parallel to the plane in which the misalignment happens, the disk has a dipole moment of its own. The unit cells of pyro-electric materials are like this. The dipole moment per unit volume of the material is called the spontaneous polarization P_s. This net dipole moment exists in the absence of an applied electric field and is equivalent to a layer of bound charge on each flat surface. Nearby free charges such as electrons or ions are attracted to the surfaces, neutralizing the charge. Imagine that conductive electrodes are then attached to the surfaces and connected through an ampmeter. If the temperature of

[8] Theophrastus (372–287 BC), successor to Aristotle, a teacher of science and author of a 10-volume history of plants which remained the most important contribution to botanical science for the following 1200 years.

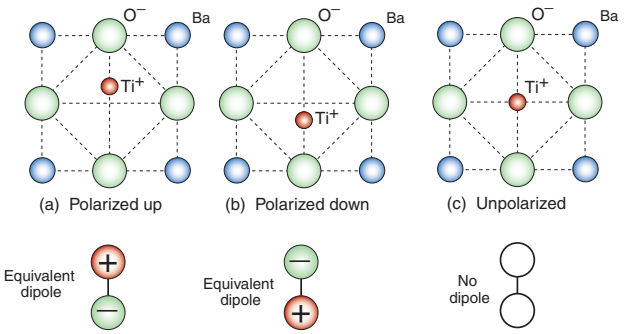


Figure 14.16 Ferro-electric materials have a permanent dipole moment that can switch: here the Ti^+ ion can flip from the upper to the lower position. Above the Curie temperature the asymmetry disappears.

the sample is constant, then so is P_s and no current flows through the circuit. An increase in temperature, however, causes expansion and that changes the net dipole moment and the polarization. Redistribution of free charges to compensate for the change in bound charge results in a current flow—the pyro-electric current—in the circuit. Cooling rather than heating reverses the sign of the current. Thus, the pyro-electric current only flows while the temperature is changing—it is the way that intruder alarms, automatic doors and safety lights are activated.

In an open circuit the free charges remain on the electrodes—and that has its uses too, described in the next section. Pyro-electric materials include minerals such as tourmaline (the one Theophrastus found), ceramics such as barium titanate, polymers such as polyvinylidene fluoride, and even biological materials, such as collagen.

Ferro-electric materials

Ferro-electrics are a special case of piezo-electric behavior. They too have an unsymmetric structure, but have the special ability to switch asymmetry. Barium titanate, $BaTiO_3$, shown schematically in Figure 14.16, is one of these. Below a critical temperature, the Curie[9] temperature (about 120°C for barium titanate), the titanium atom, instead of sitting at the center of the unit cell, is displaced up, down, to the left or to the right, as in (a) and (b). Above the Curie temperature the asymmetry disappears and with it the dipole moment, as in (c). In ferro-electrics, these dipoles spontaneously align so that large volumes of the material are polarized even when there is no applied field.

In the absence of an external field a ferro-electric divides itself up into *domains*—regions in which all the dipoles are aligned in one direction—separated

[9] Pierre Curie (1859–1906), French physicist, discoverer of the piezo-electric effect and of magnetic transformations; and husband of the yet more famous Marie Curie. He was killed in a street accident in Paris, a city with dangerous traffic even in 1906.

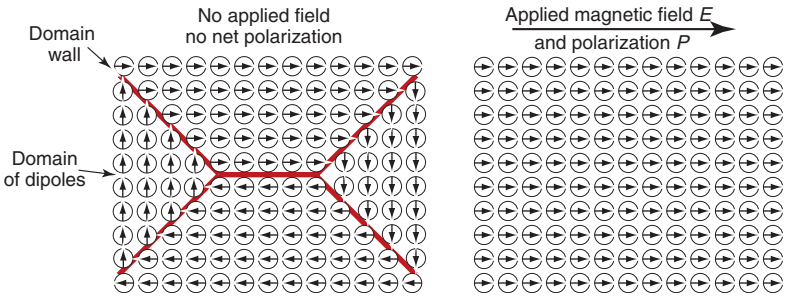


Figure 14.17 Ferro-electric domains arrange themselves so that their dipole moments cancel. In an electric field E they align, giving the crystal a net dipole moment, which can be large.

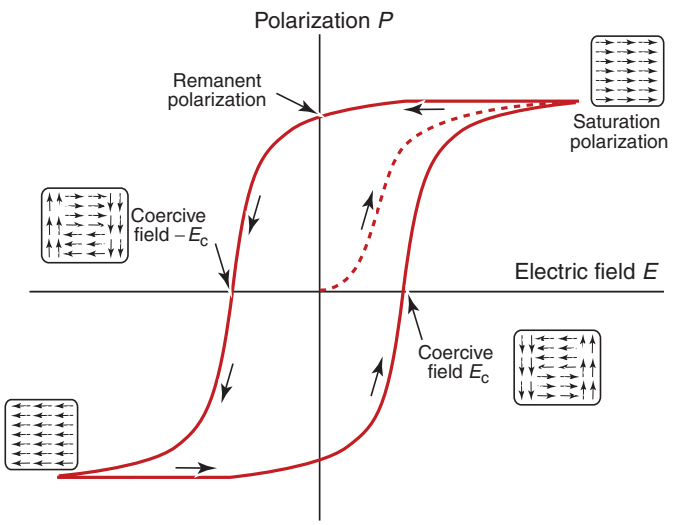


Figure 14.18 A ferro-electric hysteresis curve.

by *domain walls* at which the direction of polarization changes (Figure 14.17). The domains orient themselves so that the dipole moment of one more or less cancels those of its neighbors. If a field is applied the domain walls move so that those polarized parallel to the field grow and those polarized across or against it shrink, until the entire sample is polarized (or 'poled') in just one direction. Figure 14.18 shows how the polarization P changes as the field E is increased: P increases, reaching a maximum at the *saturation polarization*, P_s. If the field is now removed, a large part of the polarization remains (the *remanent polarization*), which is only removed by reversing the field to the value $-E_c$, the *coercive field*. The figure shows a complete cycle through full reverse polarization, ending up again with full forward poling. The little inserts show the domain structures round the cycle.